Data Flow

Food recognition

1. User uploads an image and optionally records voice or types text.
2. Frontend sends the files to the backend.
3. Backend validates type (JPG, PNG, WEBP) and size.
4. Image is resized to 256, center-cropped to 224 and normalized.
5. The PyTorch model predicts the dish and confidence.
6. The nutrition lookup returns calories, macros and a health score for that dish.
7. Frontend shows dish, confidence, serving, calories range, macros, health score.

Meal planning

1. User fills in the profile form (age, gender, height, weight, activity, goal, diet, cuisine, allergies, medical conditions, recent food).
2. Backend builds a prompt from the inputs.
3. Prompt is sent to Ollama (Llama).
4. The generated plan is returned and shown in the right panel.
5. If Ollama fails, the UI shows the error message.

Freshness, hygiene, smart kitchen

Same pattern: input from the UI, backend module processes it, result returned as JSON. TODO: add input and output details for each.